Yujia Huang

Curriculum Vitae

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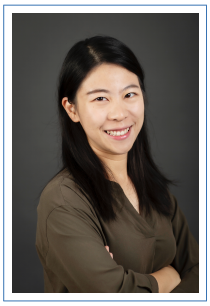

Education

2017–present **PhD, Electrical Engineering**, *California Institute of Technology*, Pasadena, USA.
Subject minor in Applied and Computational Mathematics. Advisor: Anima Anandkumar.
Research interests: Representation learning, generative models, probabilistic modeling, neuro-inspired machine learning

2013–2017 : **B.E., Opto-electronics Science and Engineering**, *Zhejiang University*, Hangzhou, China.
Minor: Advanced Honors Class of Engineering (ACEE), Chu Kochen Honors College

Publications

2020 **Yujia Huang**, James Gornet, Sihui Dai, Zhiding Yu, Tan Nguyen, Doris Tsao, and Anima Anandkumar. Neural networks with recurrent generative feedback. In *NeurIPS*, 2020.

2020 **Yujia Huang**, Michelle Cua, Joshua Brake, Yan Liu, and Changhuei Yang. Investigating ultrasound–light interaction in scattering media. *Journal of Biomedical Optics*, volume 25, page 025002. International Society for Optics and Photonics, 2020.

2020 Haowen Ruan, Yan Liu, Jian Xu, **Yujia Huang**, and Changhuei Yang. Fluorescence imaging through dynamic scattering media with speckle-encoded ultrasound-modulated light correlation. *Nature Photonics*, pages 1–6. Nature Publishing Group, 2020.

2018 Youhua Chen, Wenjie Liu, Zhimin Zhang, Cheng Zheng, **Yujia Huang**, Ruizhi Cao, Dazhao Zhu, Liang Xu, Meng Zhang, Yu-Hui Zhang, et al. Multi-color live-cell super-resolution volume imaging with multi-angle interference microscopy. *Nature communications*, volume 9, pages 1–8. Nature Publishing Group, 2018.

2017 **Yujia Huang**, Dazhao Zhu, Luhong Jin, Cuifang Kuang, Yingke Xu, and Xu Liu. Laser scanning saturated structured illumination microscopy based on phase modulation. *Optics Communications*, volume 396, pages 261–266. Elsevier, 2017.

2017 Yue Fang, **Yujia Huang**, Shaocong Liu, Cuifang Kuang, and Xu Liu. Superresolution optical microscopy. *Microscopy Methods in Nanomaterials Characterization*, pages 241–291. Elsevier, 2017.

Presentations

2020 Learning efficiently with biologically inspired feedback. In *ECCV Virtual Tutorial on New Frontiers forLearning with Limited Labels or Data*, 2020.

Research Experience

NVIDIA, Santa Clara, USA

June,2020 – present ***Large scale training for neural networks with feedback***.
Developing residual updates and prototypical representation in the CNN-F framework.
Scaling up the CNN-F model onto imagenet with adversarial training with multi-node training.

Mentor : Dr. Zhiding Yu, *Senior Research Scientist, NVIDIA*

California Institute of Technology, Pasadena, USA

August,2019 – present ***Neural networks with recurrent generative feedback***.
Developed the CNN-F framework which introduces recurrent generative feedback into neural networks.
Improved adversarial robustness and corrupted robustness of feedforward neural networks with the proposed feedback mechanism.
Proposing feedback models to better explain dynamic neuronal activities.
Exploring feedback with GANs and VAEs.

Advisor : Dr. Anima Anandkumar, *Bren Professor, Caltech*

August,2017 – August,2019 ***Imaging through scattering media***.
Investigated ultrasound-light interaction in scattering media.
Studied the ultrasound tagging efficiency in a manner consistent with its definition and experimentally verified the contributive (or noncontributive) relationship between the mechanisms involved in the ultrasound optical modulation process.

Advisor : Dr. Changhuei Yang, *Thomas G. Myers Professor, Caltech*

Awards

2020 Rising Stars in EECS (UC Berkeley)

2019 Caltech C Scholarship

2017 Top 10 Academic Achievements of Zhejiang University

2016 Chu Kochen Scholarship (highest honor for Zhejiang University undergrads)

Teaching Assistantship

Spring, 2020 : **CS165: Foundations in Machine Learning and Statistical Inference**, Caltech.

Spring, 2019 : **CS165: Foundations in Machine Learning and Statistical Inference**, Caltech.

Spring, 2018 : **EE151: Electromagnetic Engineering**, Caltech.